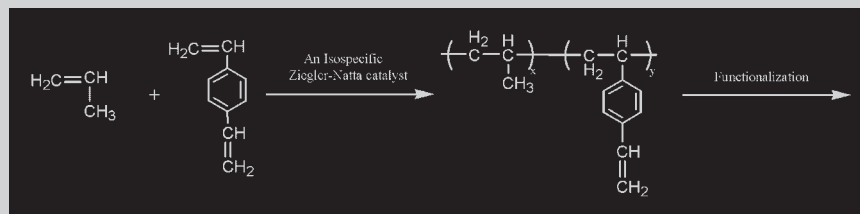


Summary: Copolymerization of propylene and 1,4-divinylbenzene was successfully performed by a MgCl_2 -supported TiCl_4 catalyst, yielding isotactic poly(propylene) (*i*-PP) polymers containing a few pendant styrene groups. With a metalation reaction with butyllithium and a hydrochlorination reaction with dry hydrogen chloride, the pendant styrene

groups were quantitatively transformed into benzyllithium and 1-chloroethylbenzene groups, respectively, which allowed the synthesis of *i*-PP-based graft copolymers by living anionic and atom transfer radical (ATRP) polymerization mechanisms.



The incorporation of styrene pendant groups into isotactic poly(propylene) using a Ziegler–Natta catalyst gave functionalized polymers able to undergo living anionic and atom transfer radical (ATRP) polymerizations.

Synthesis and Functionalization of Isotactic Poly(propylene) Containing Pendant Styrene Groups

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Introduction

For the functionalization of polyolefins, there has been tremendous interest in recent years in the design and synthesis of polyolefins containing reactive groups.^[1] This is because the direct copolymerization of an α -olefin with a functional vinyl monomer by a Ziegler–Natta or metallocene catalyst, the most straightforward way of preparing functional polyolefins, usually suffers from significantly reduced catalyst activity and polymer molecular weight because of the deactivation of catalyst by heteroatom (O, N, etc)-containing functional groups.^[2] As an alternative approach, by designing and synthesizing a polyolefin containing reactive groups that are free of catalyst-deactivating heteroatoms but very facile in functional transformation reactions, the functionalization of polyolefins by the reactive polyolefin “intermediate” is able to effectively avoid

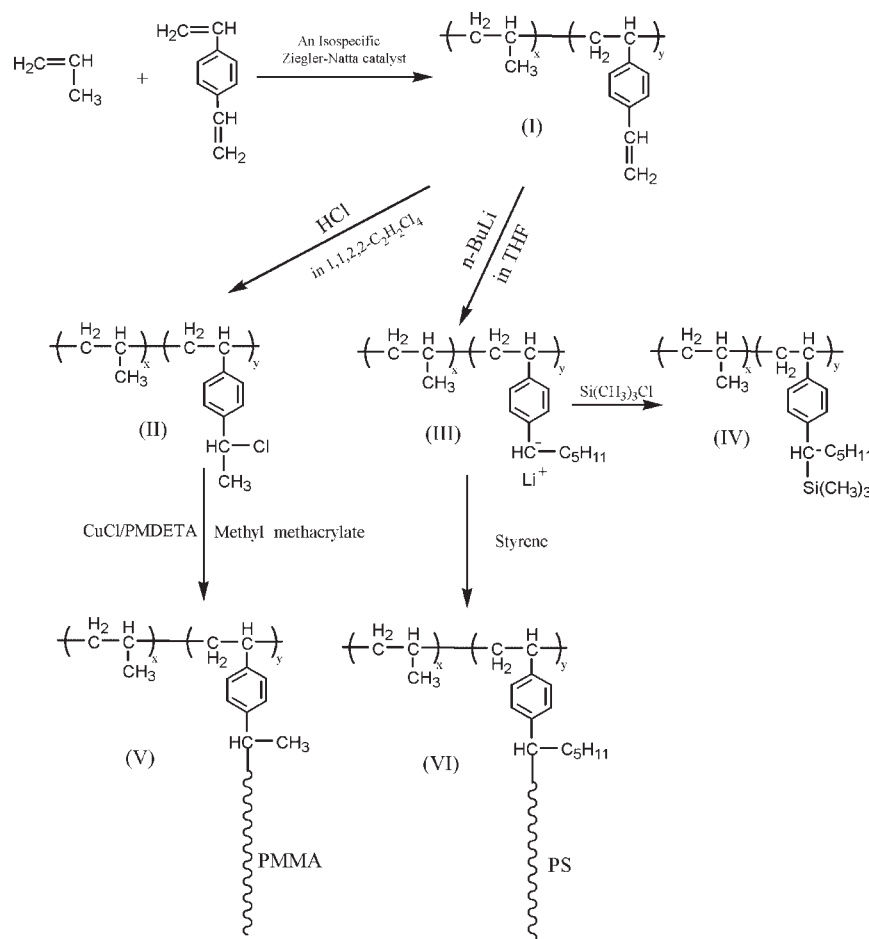
catalyst deactivation and produces functional polyolefins with well-defined molecular structures. A variety of reactive polyolefins, including polyolefins containing alkylborane groups,^[3] polyolefins containing *p*-methylstyrene units,^[4] and various unsaturated polyolefins^[5] have been synthesized using Ziegler–Natta or metallocene catalysts by the copolymerization of an α -olefin monomer with the corresponding reactive group-containing comonomers and these have been applied as “intermediates” to prepare functional polyolefins possessing versatile functionalities. We recently reported the copolymerization of ethylene^[6] and the terpolymerization of ethylene and propylene, and ethylene and oct-1-ene^[7] with 1,4-divinylbenzene using a metallocene catalyst, resulting in polyethylene and ethylene-*co*-propylene and ethylene-*co*-oct-1-ene copolymers containing pendant styrene groups. Owing to the versatile interconversion reactions involving styrene, a polyolefin

containing many pendant styrene groups was shown to be an effective “intermediate” to access a broad range of functional polyolefins. However, as we tried to incorporate 1,4-divinylbenzene units into isotactic poly(propylene) (*i*-PP), another important type of polyolefins, to prepare *i*-PP containing pendant styrene groups, we failed using metallocene catalysts. The copolymerization reactions between propylene and 1,4-divinylbenzene were unfruitful using typical isospecific metallocene catalysts including *rac*-Et(Ind)₂ZrCl₂/MAO, *rac*-Me₂Si(Ind)₂ZrCl₂/MAO, *rac*-Me₂Si(2-Me-Ind)₂ZrCl₂/MAO, and *rac*-Me₂Si(2-Me-4-Naph-Ind)₂ZrCl₂/MAO (MAO = methylaluminoxane, Ind = indenyl, Naph-Ind = naphthyl-indenyl). This phenomenon was repeated in the cases of propylene copolymerization with other styrenic monomers including styrene^[8,10] and *p*-methylstyrene^[9,10] and was reasoned by the defined yet contradictory regiospecificities of metallocene catalyst to the insertions of propylene (by a 1,2-manner) and styrene (by a 2,1-manner) that caused serious steric jamming around the active center to fatally retard the

consecutive monomer propagation during the polymerization.

Results and Discussion

Despite the unsuccessful copolymerization of propylene and 1,4-divinylbenzene using metallocene catalysts, the goal of introducing some styrene groups into *i*-PP still deserves to be pursued. Our approach to gain this goal was to resort to heterogeneous Ziegler–Natta catalysts for the copolymerization of propylene and 1,4-divinylbenzene with the idea that Ziegler–Natta-catalyzed polymerization could only proceed by a 1,2-insertion mode in both the propylene and styrene incorporations^[9] so that the copolymerization would not abort because of the steric congestion of the active sites (Scheme 1). On the other hand, because the two styrene moieties in a divinylbenzene molecule are both polymerizable in a Ziegler–Natta-catalyzed copolymerization reaction, it is also very important and interesting



Scheme 1. Copolymerization of propylene and 1,4-divinylbenzene and subsequent functionalization reactions.

to explore if the heterogeneous isospecific Ziegler–Natta catalysts are able to control the incorporation of 1,4-divinylbenzene in a selective way, that is, by reacting with only one of the two vinyl groups.

Synthesis of *i*-PP Containing Pendant Styrene Groups

The Ziegler–Natta-catalyzed copolymerization reactions between propylene and 1,4-divinylbenzene were performed in heptane solution under an atmospheric pressure of propylene using an MgCl_2 -supported TiCl_4 catalyst containing an internal electron donor (diisobutyl phthalate, DIBP) combined with AlEt_3 activator and dimethoxydiphenylsilane (DDS) external donor. The copolymerization was examined by varying the concentrations of 1,4-divinylbenzene in the copolymerization feed as well as other polymerization conditions. Table 1 summarizes the results of the copolymerization reactions.

As expected, in spite of the slightly decreased catalyst activities at higher concentrations of 1,4-divinylbenzene in the copolymerization feed, which was attributable to the lower reactivity of 1,4-divinylbenzene relative to propylene, the copolymerization reactions catalyzed by the isospecific Ziegler–Natta catalyst generally proceeded at fairly satisfactory rates, suggesting that there was no steric jamming of an active site occurring during the propagation of polymer chains. The catalyst still maintained a high activity value ($114 \text{ kg polymer} \cdot (\text{mol Ti} \cdot \text{h})^{-1}$) when a relatively high concentration of 1,4-divinylbenzene was added to the copolymerization (run no. 5 in Table 1).

All the obtained copolymers are completely soluble in common organic solvents, such as xylene, 1,1,2,2-tetrachloroethane, and 1,2,4-trichlorobenzene at elevated temperatures, which allows detailed structural analyses by using nuclear magnetic resonance (NMR) spectroscopy. Figure 1 shows the ^1H NMR spectra of three propylene-1,4-divinylbenzene copolymers containing 0.29, 0.44, and

0.65 mol-% of 1,4-divinylbenzene units. In addition to the major peaks at 0.95, 1.35, and 1.65 ppm corresponding to the CH_3 , CH_2 , and CH groups in the propylene units, several chemical shifts are observed at 2.6 ppm ($\text{CH}_2\text{CH}(\text{Ph})\text{CH}_2$), 5.2 and 5.7 ppm (doublet, $\text{CH}=\text{CH}_2$), and 6.7 ppm (doublet of doublet, $\text{CH}=\text{CH}_2$), and aromatic proton peaks at 7.1 and 7.4 ppm (C_6H_4). It is very interesting to compare the integrated intensities of these six minor peaks as they contain important information about the incorporation modes of 1,4-divinylbenzene. All the obtained copolymers show near 1:1:1:1:2:2 ratios between the peaks at 2.6, 5.2, 5.7, 6.7, 7.1, and 7.4 ppm, suggesting (i) the selective enchainment of 1,4-divinylbenzene by only one of its two double bonds, and (ii) the in-chain instead of chain-end placement of the incorporated 1,4-divinylbenzene units. Considering the generally low incorporations of 1,4-divinylbenzene in the obtained copolymers, we attribute the selective mono-enchainment of 1,4-divinylbenzene in the copolymers to the combined advantages of (i) the presence of abundant free 1,4-divinylbenzene comonomer in the copolymerization systems, and (ii) the use of Ziegler–Natta catalysts for the copolymerization that show poor capability of incorporating bulky monomers. In addition, the steric hindrance at the residual double bond of the incorporated 1,4-divinylbenzene units would be further enhanced by the adjacent polymer chain and thus effectively prevented them from further participating in the polymerization reaction.

To further confirm the linear structure of the obtained poly(propylene-*co*-1,4-divinylbenzene) copolymers, we subjected one sample to gel permeation chromatography (GPC) analysis. In Figure 2 is shown the GPC curve of a poly(propylene-*co*-1,4-divinylbenzene) copolymer containing 0.53 mol-% of incorporated 1,4-divinylbenzene units. Giving a molecular weight ($\bar{M}_n$) and molecular-weight distribution of $36\,170 \text{ g} \cdot \text{mol}^{-1}$ and 7.22, respectively, the GPC curve exhibits a smooth, unimodal signal

Table 1. A summary of copolymerization reactions between propylene and 1,4-divinylbenzene catalyzed by a MgCl_2 -supported TiCl_4 catalyst.

Run no.	Polymerization conditions ^{a)}		Yield g	Activity ^{b)}	1,4-DVB in polymer mol-%	$\bar{M}_n \times 10^{5c)}$ $\text{g} \cdot \text{mol}^{-1}$	T_m °C	χ_c %
	1,4-DVB	T						
	$\text{mol} \cdot \text{L}^{-1}$	°C						
1	0	50	8.26	236	0	1.21	159.2	47.5
2	0.058	50	7.14	204	0.29	1.89	156.7	39.4
3	0.154	50	5.36	153	0.53	1.47	154.6	38.8
4	0.308	50	4.30	123	0.56	1.30	153.9	34.7
5	0.615	50	3.99	114	0.65	0.86	152.6	31.1
6	0.154	30	6.30	180	0.44	2.25	157.1	38.3
7	0.154	70	1.82	52	0.49	1.22	153.1	38.4

^{a)} Other conditions: atmospheric pressure of propylene, 50 mL heptane, $\text{MgCl}_2/\text{DIBP}/\text{TiCl}_4$ catalyst, $[\text{Ti}] = 0.07 \text{ mmol}$, AlEt_3 cocatalyst, DDS external donor, $[\text{Al}]/[\text{Ti}] = 100$, $[\text{DDS}]/[\text{Ti}] = 10/1$, polymerization time = 30 min.

^{b)} Catalyst activity: kilograms of polymer per mole of Ti per hour.

^{c)} The viscosity-averaged molecular weight was calculated according to the equation of $[\eta] = 1.10 \times 10^{-4} \bar{M}_n^{0.8}$.^[14]

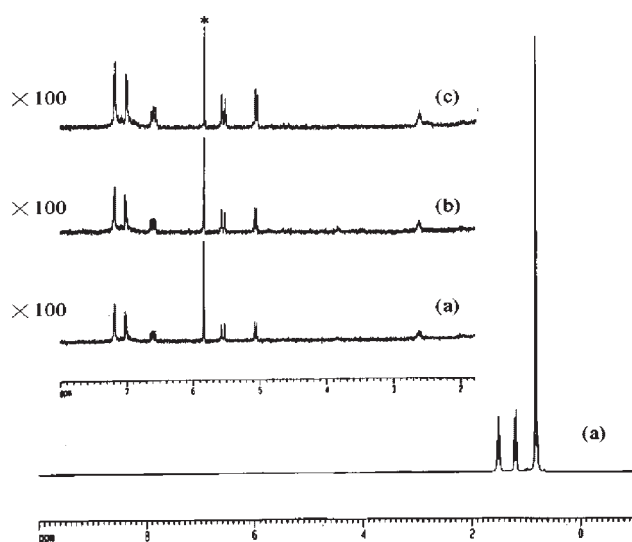


Figure 1. ^1H NMR spectra of three poly(propylene-*co*-1,4-divinylbenzene) copolymers containing (a) 0.29, (b) 0.44, and (c) 0.65 mol-% of 1,4-divinylbenzene units. (Solvent: 1,1,2,2-(D_2)tetrachloroethane; $T = 110^\circ\text{C}$, the asterisk represents solvent).

and shows no sign of the existence of cross-linked polymers.

The 1,4-divinylbenzene contents as well as styrene group concentrations of the copolymers were calculated based on the comparison of the integration of any one of the peaks at 5.2, 5.7, or 6.7 ppm with that of the sum of peaks at 0.95, 1.35, and 1.65 ppm, together with consideration of the number of protons each chemical shift represents. Despite the rapid increase of the concentrations of 1,4-divinylbenzene in the copolymerization feeds, its incorporation into the copolymers increases very slowly because of the large reactivity difference between propylene and the bulky styrenic monomer. As such, adjusting the polymerization conditions did not significantly affect the comonomer incorporation. The maximum incorporation of 1,4-divinyl-

benzene in the copolymer under our current conditions is lower than 1 mol-%.

In general, the copolymerization results of propylene and 1,4-divinylbenzene involving catalyst activities and comonomer incorporations using the MgCl_2 -supported TiCl_4 catalyst are consistent with those obtained by Soga and Yanagihara,^[11] Lin and Gao,^[12] and Chung and Lu^[9] in the cases of propylene copolymerization with other styrenic monomers including styrene and *p*-methylstyrene by heterogeneous Ziegler–Natta catalysts. Despite the satisfactory catalyst activities, incorporation of the bulky styrenic comonomers is usually very difficult, so unable to reach high values, which is attributable to the generally poor copolymerization capability of heterogeneous Ziegler–Natta catalysts. However, the low comonomer incorporation into the *i*-PP copolymers is propitious to preserve the original desired properties of *i*-PP. As shown in Table 1, both the molecular weights and melting temperatures of the obtained poly(propylene-*co*-1,4-divinylbenzene) copolymers show only slight decreases as compared with those of the propylene homopolymer.

Functionalization of *i*-PP Containing Pendant Styrene Groups

The limited incorporation of 1,4-divinylbenzene using the Ziegler–Natta catalyst resulted in low concentrations of reactive groups in the *i*-PP polymers, which posed a great challenge for the subsequent functionalization reactions. However, thanks to the high reactivity of styrene groups in the interconversion reactions into the initiating groups for living anionic^[7] and atom transfer radical (ATRP)^[13] polymerizations, the *i*-PP containing few pendant styrene groups was still able to serve as a reactive “intermediate” to access functional *i*-PP with controlled amounts of functional groups by synthesizing *i*-PP graft copolymers. Both the metalation reaction by butyllithium and the hydrochlorination reaction by anhydrous HCl of the poly(propylene-*co*-1,4-divinylbenzene) copolymers could reach quantitative levels under mild reaction conditions, which conveniently provided polymeric benzyl lithium and polymeric benzyl chloride initiating sites for the anionic and ATRP graft-from polymerizations of polar monomers (Scheme 1). As an example, in Figure 3 are shown the ^1H NMR spectra of the metalated and subsequently silylated (Figure 3a) and hydrochlorinated (Figure 3b) products of poly(propylene-*co*-1,4-divinylbenzene) copolymer containing 0.44 mol-% of 1,4-divinylbenzene units (run 6 in Table 1, pendant styrene group concentration is also 0.44 mol-%). The starting poly(propylene-*co*-1,4-divinylbenzene) copolymer has a viscosity-averaged molecular weight of $225\text{ kg}\cdot\text{mol}^{-1}$, a melting point of 157.1°C , and a crystallinity of 38.3%. While the hydrochlorination reaction was performed by bubbling HCl gas into a suspension of the polymer in 1,1,2,2-tetrachloroethane at 80°C , the

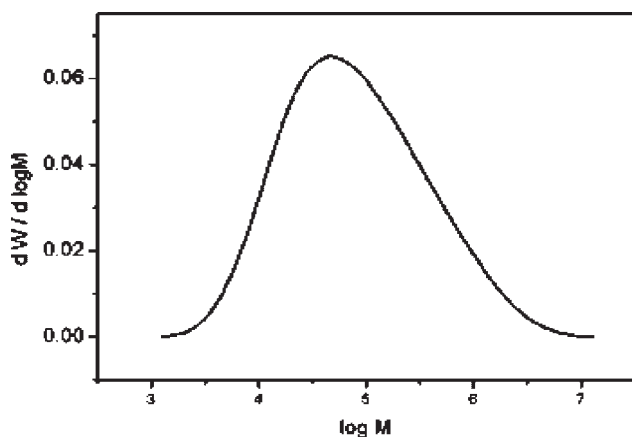


Figure 2. GPC trace of a poly(propylene-*co*-1,4-divinylbenzene) copolymer containing 0.53 mol-% of 1,4-divinylbenzene units (run 3 in Table 1).

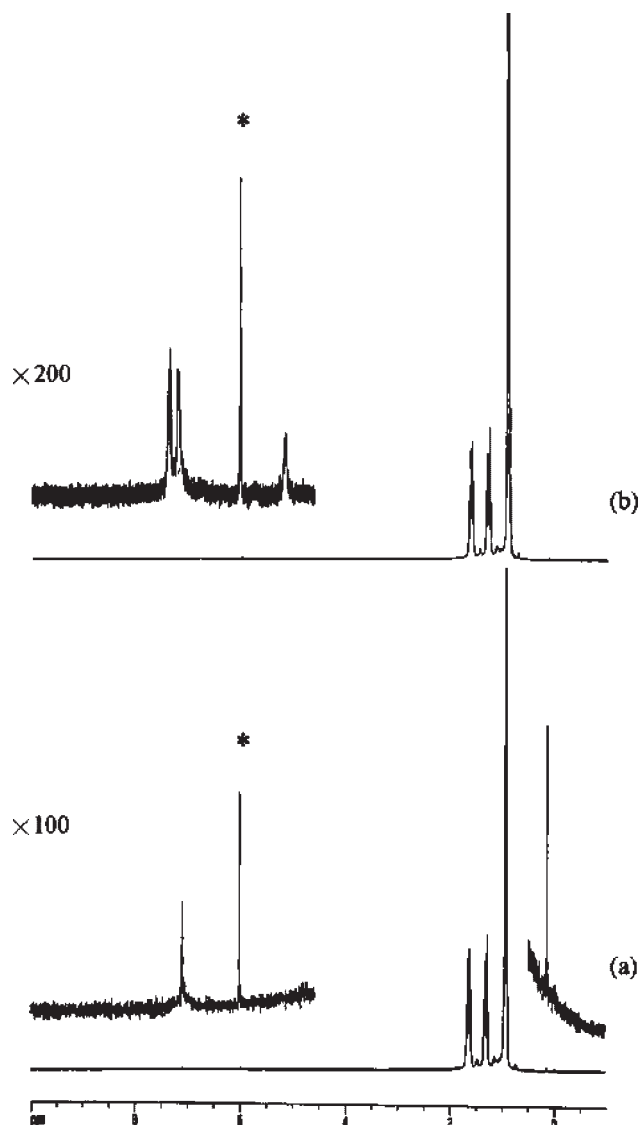


Figure 3. ^1H NMR spectra of a poly(propylene-*co*-1,4-divinylbenzene) copolymer containing 0.44 mol-% of 1,4-divinylbenzene units after (a) metalation with BuLi and subsequent silylation with trimethylsilyl chloride, and (b) hydrochlorination with HCl. (Solvent: 1,1,2,2-(D_2)tetrachloroethane; $T = 110^\circ\text{C}$).

metalation reaction was carried out in cyclohexane at 40°C using butyllithium as the metalating reagent followed by a silylation reaction with trimethylsilyl chloride to quantify the efficiency of the metalation reaction (Scheme 1). As shown in Figure 3a and 3b, all the peaks corresponding to the styrenic vinyl groups at 5.2, 5.7, and 6.7 ppm completely disappear after metalation and hydrochlorination. The new peaks at 0.05 ppm in Figure 3a and at 5.1 ppm in Figure 3b correspond to the $-\text{Si}(\text{CH}_3)_3$ and $-(\text{C}_6\text{H}_4)\text{CHCl}-$ groups, respectively. Interestingly, while the aromatic protons still show two separated peaks in the spectrum of the hydrochlorinated product, as they do for the starting poly(propylene-*co*-1,4-divinylbenzene) copolymer, because of the presence of chlorine at the benzylic position, they exhibit

only one single peak in the spectrum of the metalated and subsequently silylated product because of the similar chemical environments around them. The experimental results indicate clean reactions and almost quantitative conversions in each of the metallation, silylation, and hydrochlorination reactions.

The controllable introduction of polar functional groups into *i*-PP through the styrene group-containing “intermediate” was embodied in the preparation of *i*-PP-*g*-PS (polystyrene) and *i*-PP-*g*-PMMA (poly(methyl methacrylate)) graft copolymers. The above-obtained metalated and hydrochlorinated poly(propylene-*co*-1,4-divinylbenzene) copolymers were used to initiate the graft-from polymerization of styrene and methyl methacrylate (MMA) by living anionic and ATRP (in the presence of CuCl and pentamethyldiethylenetriamine) mechanisms, respectively. By controlling the monomer conversions during the graft polymerization reactions, functional *i*-PP polymers containing defined yet adjustable amounts of styrene and MMA units were achieved. Figure 4 shows ^1H NMR spectra of two anionically prepared *i*-PP-*g*-PS graft copolymers

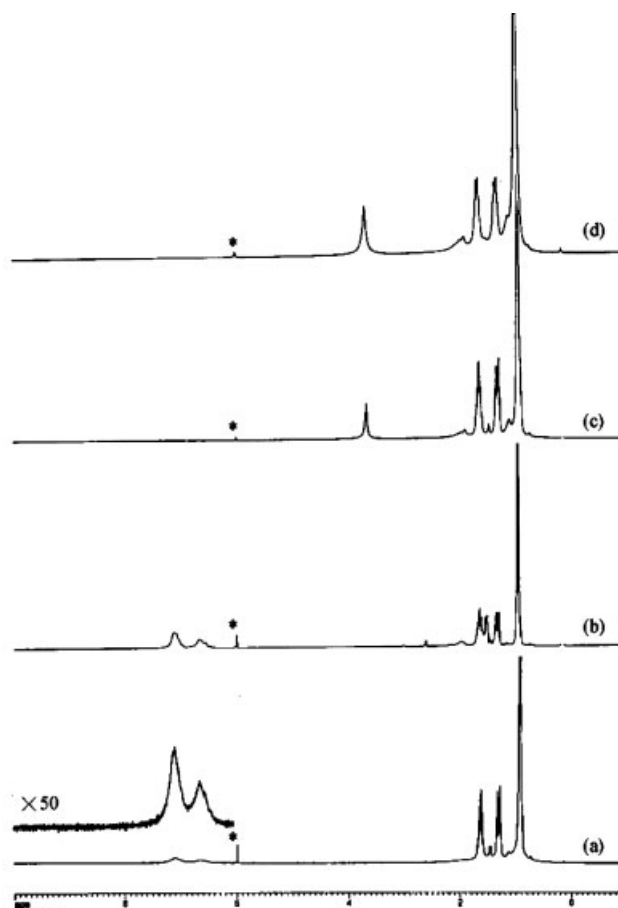


Figure 4. ^1H NMR spectra comparison between two anionically prepared *i*-PP-*g*-PS graft copolymers containing (a) 18.0 and (b) 46.0 wt.% of styrene units and two radically prepared *i*-PP-*g*-PMMA graft copolymers containing (c) 22.4 and (d) 37.3 wt.% of MMA units (solvent: 1,1,2,2-(D_2)tetrachloroethane; $T = 110^\circ\text{C}$).

containing 18.0 (Figure 4a) and 46.0 wt.-% (Figure 4b) of styrene units, and two radically-prepared *i*-PP-*g*-PMMA graft copolymers containing 22.4 (Figure 4c) and 37.3 wt.-% (Figure 4d) of MMA units, respectively. All the four graft copolymers were derived from the poly(propylene-*co*-1,4-divinylbenzene) copolymer containing 0.44 mol.-% of pendant styrene groups and were obtained by simply varying the monomer conversions during the respective graft polymerization reactions. In addition to the peaks at 0.95, 1.35, and 1.65 ppm corresponding to the three types of protons in the *i*-PP backbone, the two *i*-PP-*g*-PS graft copolymers show aromatic proton peaks at 6.4–7.3 ppm in their spectra, corresponding to consecutively arrayed styrene units which are indicative of PS graft segments. On the other hand, the two *i*-PP-*g*-PMMA graft copolymers exhibit the characteristic resonance of $-\text{COOCH}_3$ in the MMA unit at 3.68 ppm, the integration of which indicates the polymeric form of MMA units.

Despite the high second polymer contents, the *i*-PP-based graft copolymers still possess high melting temperatures, which must be attributed to their well-preserved PP main-chain structure arising from the relatively low 1,4-divinylbenzene incorporation in the starting poly(propylene-*co*-1,4-divinylbenzene) copolymer, which resulted in low graft densities. Figure 5 compares DSC curves of the *i*-PP-*g*-PMMA graft copolymer containing 37.3 wt.-% of MMA units and *i*-PP-*g*-PS containing 46.0 wt.-% of styrene units with that of the starting poly(propylene-*co*-1,4-divinylbenzene) copolymer having 0.44 mol.-% of 1,4-divinylbenzene incorporation. In addition to the glass transitions of the PMMA and PS grafts observed at 101.9 and 99.1 °C, the *i*-PP-*g*-PMMA and *i*-PP-*g*-PS graft copolymers exhibit their melting points at 158.4 and 156.9 °C,

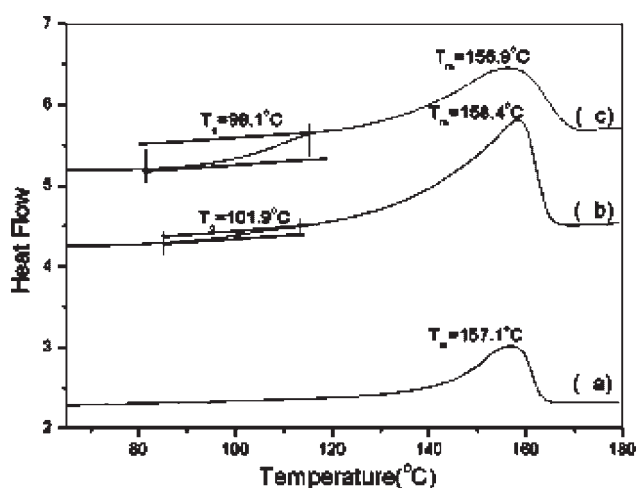


Figure 5. A comparison of DSC curves between (a) the poly(propylene-*co*-1,4-divinylbenzene) copolymer containing 0.44 mol.-% of 1,4-divinylbenzene units, (b) the *i*-PP-*g*-PMMA graft copolymer containing 37.3 wt.-% of MMA units, and (c) the *i*-PP-*g*-PS graft copolymer containing 46.0 wt.-% of styrene units.

respectively, which are comparable with that of the starting poly(propylene-*co*-1,4-divinylbenzene) copolymer.

Conclusion

In summary, the experimental results clearly show the successful preparation of *i*-PP containing pendant styrene groups by the Ziegler–Natta-catalyzed copolymerization of propylene and 1,4-divinylbenzene. The formed poly(propylene-*co*-1,4-divinylbenzene) copolymers are completely soluble in common organic solvents at elevated temperatures and exhibit high molecular weights, melting temperatures, and crystallinities. Despite the limited incorporation of 1,4-divinylbenzene units using the Ziegler–Natta catalyst, the *i*-PP polymers containing few pendant styrene groups are still effective “intermediates” for the functionalization of *i*-PP because of the quantitative inter-conversion reactions of styrene groups to living anionic and ATRP initiating sites. *i*-PP polymers containing controllable amounts of styrene or MMA units were achieved by living anionic and ATRP graft-from polymerizations, respectively.

Experimental Part

Materials and Instruments

All O₂ and moisture sensitive manipulations were carried out inside an argon-filled vacuum atmosphere dry-box equipped with a dry train. Pure 1,4-divinylbenzene was isolated from a commercially available divinylbenzene mixture (Aldrich Chemical Co.). Styrene (St, 99%), and methyl methacrylate (MMA, 99%) were vacuum distilled and stored under argon. Anisole (99%) was deoxygenated by argon purge before refluxing for 24 h and then distilling over CaH₂. CP grade heptane was deoxygenated by argon purge before refluxing for 48 h and then distilling over sodium. AlEt₃ (1.8 M in toluene), dimethoxydiphenylsilane (DDS), CuCl (98%), and pentamethyldiethylenetriamine (PMDETA, 99%) were from Aldrich and used without further purification. Polymerization grade propylene was supplied by Yansan Petrochemical Co. of China. The MgCl₂/diisobutyl phthalate (DIBP)/TiCl₄ catalyst was kindly donated by Xiangyang Chemical Co. of China with a Ti content of 2.5 wt.-%.

All ¹H NMR spectra were recorded on a Bruker AM-300 instrument in 1,1,2,2-(D₂)tetrachloroethane at 115 °C. The melting and glass transition temperatures as well as the crystallinities of the polymers were measured by differential scanning calorimeter (DSC) using a Perkin–Elmer DSC-7 instrument controller operating from 30 to 180 °C with a heating rate of 20 °C · min^{−1}. The data were collected on the second heating cycle. The intrinsic viscosity [η] of the polymers was measured in decahydronaphthalene at 135 ± 0.1 °C. Gel permeation chromatography (GPC) measurements were performed using a Waters Alliance GPC 2000V instrument operated at 150 °C with 1,2,4-trichlorobenzene as the eluent to determine polymer molecular weight and molecular-weight distribution.

Copolymerization of Propylene and 1,4-Divinylbenzene by the MgCl_2 -Supported Ziegler–Natta Catalyst

In a typical reaction (run no. 3 in Table 1), 50 mL of heptane was added into a 250-mL three-necked glass reactor equipped with a magnetic stirrer. The reactor was then saturated with atmospheric pressure of propylene at 50 °C. The compounds 1,4-divinylbenzene (1 g), AlEt_3 (3.88 mL, 1.8 M in toluene), and DDS (4.13 mL, 0.7 mmol) were subsequently injected with a syringe (Al/Si molar ratio of 10/1). The polymerization was started by adding 0.1 g of $\text{MgCl}_2/\text{DIBP}/\text{TiCl}_4$ catalyst into the reactor. After 30 min of reaction, 10 mL of acidic methanol was added to terminate the reaction. The product was collected by filtration and repeatedly washed with methanol, tetrahydrofuran (THF), and water and dried at room temperature under high vacuum for 24 h to obtain 5.36 g of white polymer powder.

Lithiation Reaction of Poly(propylene-co-1,4-divinylbenzene)

In an argon filled drybox, 3 g of poly(propylene-co-1,4-divinylbenzene) copolymer containing 0.44 mol-% of divinylbenzene units was suspended in 50 mL of anhydrous cyclohexane in a 100-mL flask equipped with a magnetic stirrer. BuLi in hexane (1 mL of 1.6 M) and 1 mL (6.5 mmol) of TMEDA were then added to the flask. The flask was then brought out of the drybox and heated to 40 °C. The lithiation reaction was allowed to proceed for 4 h before the reaction mixture was filtered and repeatedly washed with cyclohexane. A yellow polymer powder was obtained after drying under vacuum at 60 °C for 4 h (lithiated poly(propylene-co-1,4-divinylbenzene)).

Silylation Reaction of Lithiated Poly(propylene-co-1,4-divinylbenzene)

The above lithiated poly(propylene-co-1,4-divinylbenzene) (1.0 g) was suspended in 30 mL of dry cyclohexane, to which 2 mL of Me_3SiCl was added using a syringe. The reaction mixture was stirred at 40 °C for 2 h. After filtration and repeatedly washing with THF, methanol, and water, the solid polymer was collected and dried under vacuum at 60 °C for 8 h. The ^1H NMR spectrum shows a strong peak at 0.05 ppm corresponding to the methyl proton adjacent to Si.

Hydrochlorination of Poly(propylene-co-1,4-divinylbenzene)

In a 250-mL round-bottomed flask equipped with a magnetic stirrer, 2.0 g of poly(propylene-co-1,4-divinylbenzene) polymer containing 0.44 mol-% of 1,4-divinylbenzene units was suspended in 100 mL of 1,1,2,2-tetrachloroethane at 80 °C. Dry hydrogen chloride (HCl) that was generated by adding hydrochloric acid (36%) dropwise into concentrated sulfuric acid was bubbled into the suspension. After 6 h, the suspension was poured into 200 mL of ethanol, the solid was filtered off and washed with ethanol several times. The final product was dried at room temperature under high vacuum for 24 h to obtain white polymer powder.

Grafting of Styrene from Poly(propylene-co-1,4-divinylbenzene)

In one experiment (Figure 4a), 0.26 g of the above-obtained lithiated poly(propylene-co-1,4-divinylbenzene) copolymer was suspended in 50 mL of anhydrous cyclohexane, to which 4 mL of styrene was added at room temperature using a syringe. After stirring for 30 min at room temperature, 10 mL of methanol was added to terminate the graft reaction. The precipitated polymer was filtered and dried under vacuum to give 0.32 g of *i*-PP-g-PS, which accounts for about 18.0 wt.-% of polystyrene in the graft copolymer. A Soxhlet extraction experiment performed on the *i*-PP-g-PS graft copolymer using THF resulted in no soluble polymer, indicating there was no homo-polystyrene polymer formed during the graft-from reaction.

In another experiment (Figure 4b), with the same amounts of monomer and initiator, the graft reaction was prolonged for 1 h and generated 0.46 g of *i*-PP-g-PS containing 46.0 wt.-% of polystyrene segments.

Grafting of MMA from Hydrochlorinated Poly(propylene-co-1,4-divinylbenzene)

In one experiment (Figure 4c), a dry round-bottomed flask was charged with anisole (10 mL), CuCl (0.043 g, 0.3 mmol), PMDETA (0.051 g, 0.3 mmol), MMA (3 g, 30 mmol), and the above-obtained hydrochlorinated poly(propylene-co-1,4-divinylbenzene) (0.15 g). The flask was sealed and filled with purified argon. After the mixture was allowed to stir at ambient temperature for 5 min, the flask was placed in an oil bath at 95 °C for 40 min. The reaction was terminated by pouring the contents of the flask into a large amount of acidic methanol. The precipitated polymer was washed and dried under vacuum to give 0.19 g of PP-g-PMMA graft copolymer containing 22.4 wt.-% of PMMA segments. A Soxhlet extraction experiment performed on the graft copolymer using acetone resulted in no soluble polymer, indicating there was no homo-PMMA polymer formed during the graft-from reaction.

In another experiment (Figure 4d), with the same amounts of solvent, monomer, catalyst, ligand, and initiator, the graft reaction was prolonged for 1.5 h and generated 0.24 g of *i*-PP-g-PMMA containing 37.3 wt.-% of polystyrene segments.

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